Self-Adaptive Global Feature Fusion Network With Spectral Prompt for Hyperspectral Image Classification

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***Abstract*— Nowadays, foundation models have demonstrated exceptional performance across numerous downstream tasks. However, the effective application of these models to hyperspectral image classification (HSIC) is challenged by the unique characteristics of hyperspectral data, including high dimensionality, high variability, and high spatial structure complexity. Therefore, methods need to be developed, which leverage the advantages of foundation models while addressing these challenges. First, a novel HSIC algorithm based on a frozen-parameter segment anything model (SAM) encoder, called SAGFFNet, is proposed. This framework represents the first attempt to use a frozen SAM encoder for global feature extraction and to use spectral dimension data as prompts, enabling precise global spatial–spectral feature extraction with the aid of spectral information. Second, by introducing the self-adaptive padding mechanism and the global feature extraction subnetwork (GFEsNet), the model is enabled to extract distinctive and discriminative features for each category from hyperspectral data through varying padding sizes, thereby enhancing the feature extraction and generalization capabilities of the foundation model. Subsequently, the spectral feature prompt subnetwork (SFPsNet) is designed to extract spectral feature information from samples of different classes as prompt features, assisting the framework in better understanding the global features extracted by GFEsNet. Finally, the semantic information decoder subnetwork (SIDsNet) is introduced as a semantic information decoder, achieving efficient fusion of global spatial–spectral features and spectral prompt features, which significantly improves classification performance. Experiments conducted on four hyperspectral image datasets show that the proposed method outperforms nine existing approaches in terms of classification accuracy.**

***Index Terms*— Foundation model, global feature extraction, hyperspectral image classification (HSIC), segment anything model (SAM).**

I. Introduction

HYPERSPECYTAL image classification (HSIC), based on the spectral curve information of pixels, assigns a land cover category to each pixel, serving as a critical segment and component in the application of HSI [1], [2] [3]. This technology has demonstrated immense potential in various fields such as intelligent agriculture, environmental monitoring, geological surveying, and military reconnaissance [4], [5] [6].

With the continuous advancement of artificial intelligence technology, deep learning algorithms such as CNN [7], [8], GCN [9], [10], and transformer [11], [12] are capable of automatically learning complex relationships and patterns within data and have become mainstream algorithms for HSIC [13], [14]. For instance, Wu et al. [15] proposed a multimodal remote sensing image classification network by introducing a cross-channel reconstruction module, which makes the features of different modalities fused in a more compact way. In another approach, Hong et al. [16] developed SpectralFormer, which is designed to learn local spectral sequence information from adjacent bands. In addition, Hong et al. [17] proposed a subpixel-level framework utilizing the intrinsic properties of hyperspectral–multispectral images effectively for the fusion task, which is more applicable to various real remote sensing images tasks.

However, in practical applications, acquiring labeled samples is costly, often resulting in insufficient or completely unlabeled samples for newly acquired hyperspectral images [18]. To address the challenge of limited labeled samples, many small-sample deep learning algorithms and transfer learning methods have been proposed [19]. For instance, Fang et al. [20] designed CARGNet for point-supervised change detection, which can effectively address the challenge of limited availability of labeled samples. Xu et al. [21] introduced a change capsule network utilizing vector cosine and vector difference for feature comparison, achieving strong performance in remote sensing image change detection with limited training data.

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A. Main Challenge

In summary, with the continuous advancement of artificial intelligence technology, deep learning algorithms have the capability to learn the differences in hyperspectral data categories [22]. However, despite the ability of these deep learning models with relatively few parameters to handle classification tasks for specific scenes or datasets after targeted training, these methods still face challenges in representing hyperspectral information and generalization [23]. Specifically, these approaches exhibit strong intradomain representation capabilities but weak interdomain generalization capabilities. When a trained model is used to classify datasets from different sensors or scenes [24], its performance often falls short of application requirements. In such cases, the pretrained model needs to be retrained on the new scene data until satisfactory classification performance is achieved. This severely restricts the application of hyperspectral data and becomes a bottleneck in its value chain.

Surprisingly, in the past two years, with the advent of massive datasets and powerful computational resources, foundation models have made breakthroughs in image understanding, such as vision transformer [25], masked autoencoders (MAEs) [26], and segment anything model (SAM) [27]. Their main advantages include: 1) strong feature expression capability that is more conducive to learning the intrinsic features of data and 2) large network models with many parameters, leading to stronger theoretical generalization ability.

B. Potential Capability

Therefore, given the aforementioned advantages of foundation models, leveraging them for remote sensing tasks holds significant potential and promises new solutions and opportunities. For instance, the SAM model demonstrates strong feature extraction capabilities in the domain of remote sensing image segmentation by accurately capturing critical features such as edges, textures, and colors [28]. This indicates that the SAM model has impressive potential in the field of HSIC.

More specifically, the foundation model applied to the field of spectral-image classification has attracted increasing interest. Zhang et al. [29] introduced the language-aware domain generalization network, proposing an algorithm that, when exclusively trained on the source domain, can successfully perform cross-domain classification tasks for hyperspectral images. In response to the limitations of existing datasets characterized by small-scale samples, limited class categories, and a lack of diversity, Sun et al. [30] presented a large-scale HSI dataset. This dataset comprises 652 743 samples distributed across 20 distinct categories, aiming to address the challenges posed by the scarcity of diverse and abundant hyperspectral data for classification tasks. He et al. [31] introduced a foundation model adaptation (FMA) framework for the classification of multimodal remote sensing data encompassing hyperspectral data that eliminate the need for fine-tuning parameters.

Results demonstrate that the FMA framework outperforms advanced methods in multimodal remote sensing data classification. Hong et al. [32] have pioneered the development of a universal remote sensing foundation model, named SpectralGPT, which validates the substantial advantages and potential of foundation models in the realm of spectral remote sensing data. Huang et al. [33] constructed the foundation model-based spectral–spatial transformer to address generalized zero-shot classification, which opens a new window for HSI classification tasks, especially for HSI zero-shot classification.

C. Practical Approach

Based on the aforementioned work, it can be summarized that, currently, there are two practical approaches to utilize foundation models for the HSIC task. One approach involves constructing a new foundation model and retraining its network parameters using a large amount of hyperspectral data. Importantly, the recent research finding by Wang et al. [34] from Wuhan University has been preprinted, introducing HyperSIGMA, an advanced hyperspectral intelligent understanding foundation model. HyperSIGMA represents the first billion-scale model specifically tailored for interpreting hyperspectral images, offering a unified solution for tasks at both high and low levels. In addition, they curated the HyperGlobal-450K dataset, a global-scale collection of hyperspectral images surpassing existing multispectral and hyperspectral datasets in size. This dataset is poised to advance large-scale pretraining efforts in the field of hyperspectral foundation models. However, due to the complex network structures, vast amounts of data, and parameters involved, this method incurs expensive implementation costs and high practical barriers, making it challenging for ordinary researchers to participate substantially.

The second approach involves directly or fine-tuning pretrained foundation models to perform HSIC tasks. This significantly lowers the research threshold while maximizing the advantages of foundation models. As a highly practical method, this approach may become a focal point of future research. Note that research on the foundation model for HSIC is still in its early stages, with several fundamental issues yet to be resolved. Hyperspectral data, characterized by high spectral dimensions, spectral variability, and complex spatial structures, exhibit significant differences compared to natural images in terms of land cover types, spatial resolution, and imaging conditions. Pretrained foundation models struggle to adapt well to hyperspectral data, encountering unique challenges.

Thus, on the basis of the above analysis, this article explores a second avenue of research, i.e., introducing a pretrained foundation model and adapting it to hyperspectral data. To the best of our knowledge, this is the first attempt to introduce a pretrained SAM model into the field of HSIC.

To be specific, this article proposes an algorithm, namely, SAGFFNet based on the pretrained SAM encoder; it consists of a global feature extraction subnetwork (GFEsNet), a spectral feature prompt subnetwork (SFPsNet), and a semantic information decoder subnetwork (SIDsNet). The flowchart of our proposed SAGFFNet is shown in Fig. 1. SAGFFNet leverages GFEsNet to extract global spatial–spectral information and utilizes SFPsNet to extract 1-D spectral information. These

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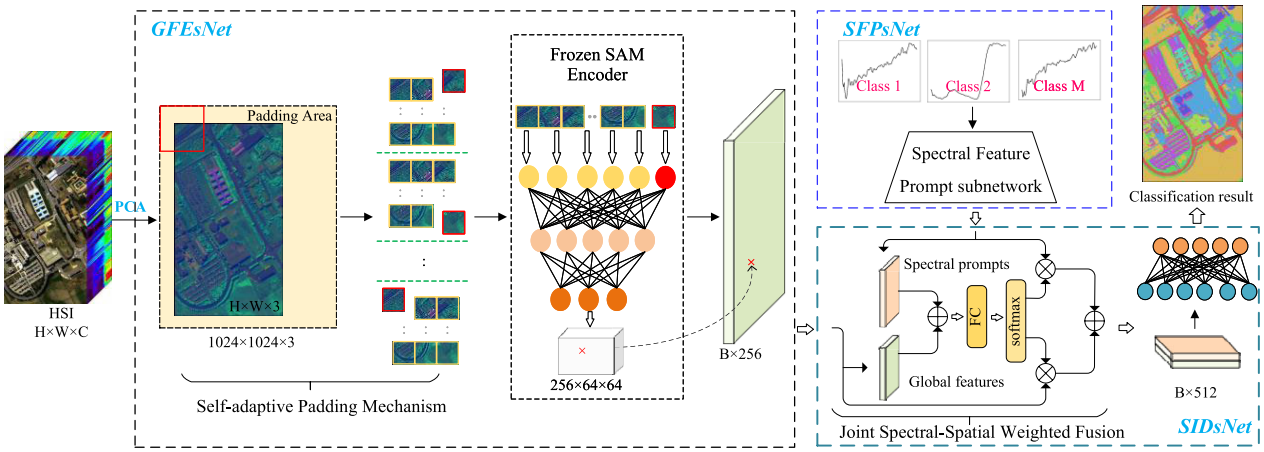


Fig. 1. Flowchart of the SAGFFNet framework, where H, W, and C represent the length, width, and dimensionality of the input HSI, respectively, B represents the batch size, M denotes the number of categories, $\bigoplus$ signifies the feature concatenation, and $\bigotimes$ means the weighted multiplication. The detailed network structure of the SFPsNet will be provided in Section II-B.

two types of features are then concatenated and fed into SIDsNet to accomplish pixel-level classification.

The main contributions of this article can be summarized as follows.

1) First, a novel SAGFFNet framework is proposed. This is the first attempt to use a frozen SAM encoder to extract global features and employ spectral dimension data as a prompt, enabling accurate global spatial–spectral feature extraction with the aid of spectral information. In addition, this framework explores the mechanism of adapting the SAM model to hyperspectral data.
2) Second, the self-adaptive padding mechanism and the GFEsNet are introduced to the foundation model, which aims at fully leveraging the strong feature extraction and generalization capabilities of the foundation model. By padding with different sizes, this mechanism enables the model to extract distinctive and discriminative features for each category from the hyperspectral data.
3) Third, the SFPsNet is designed to extract spectral feature information from samples of different classes as the prompt feature, aiding the overall framework in better understanding the global features extracted by GFEsNet. In addition, SIDsNet is introduced as a semantic information decoder, achieving efficient fusion of the global spatial–spectral feature and the spectral prompt feature, further enhancing the classification performance of the SAGFFNet framework.

The structure of this article is given as follows. Section II describes the methodology of the proposed SAGFFNet framework, detailing its components: GFEsNet, SFPsNet, and SIDsNet. Section III presents the experimental results and parameter analysis. Next, Section IV discusses the advantages and disadvantages of the SAGFFNet in comparison with existing approaches. Finally, Section V provides a summary of this article.

II. Proposed Method

The flowchart of the SAGFFNet algorithm is shown in Fig. 1. First, the HSI is input into the GFEsNet to obtain global spatial–spectral features. Next, the SFPsNet is employed to extract 1-D spectral information from different class samples, serving as the prompt feature. These two types of features are then concatenated and fed into the SIDsNet, which efficiently fuses the global spatial–spectral features and the spectral prompt features to achieve the final pixel-level classification results.

A. GFEsNet

To fully leverage the strong feature extraction and generalization capabilities of the foundation model, this article introduces the self-adaptive padding mechanism and subsequently designs GFEsNet. The self-adaptive padding mechanism can adaptively pad HSIs to extract global spatial–spectral information. In addition, the design philosophy of GFEsNet is to utilize the pretrained SAM encoder to effectively capture complex global features in the data without fine-tuning the encoder parameters, thereby enhancing the overall performance of the SAGFFNet framework.

In GFEsNet, when extracting features for a particular labeled sample, a global feature extraction strategy is employed, expanding the receptive field to encompass all patches, thus achieving interaction with global information. This approach is constructed because the spatial resolution of hyperspectral data is relatively low, and significant spectral variations can occur between the same ground objects at different locations. By utilizing an attention mechanism to implement a weighted fusion of spectral information from different distances into the feature of a specific pixel, the accuracy of the features can be effectively enhanced.

Specifically, the self-adaptive padding mechanism and GFEsNet are introduced as follows, and the self-adaptive padding mechanism is shown in Fig. 2.

For the sample points (x, y), the size of padding is assumed to be p_1 and p_2, as shown in Fig. 2(b). To ensure that all sample pixels in the hyperspectral data can become the central pixels of the padded patches under different padding values, it is necessary to adaptively adjust the padding size. The relationship between the coordinates of the central pixels and

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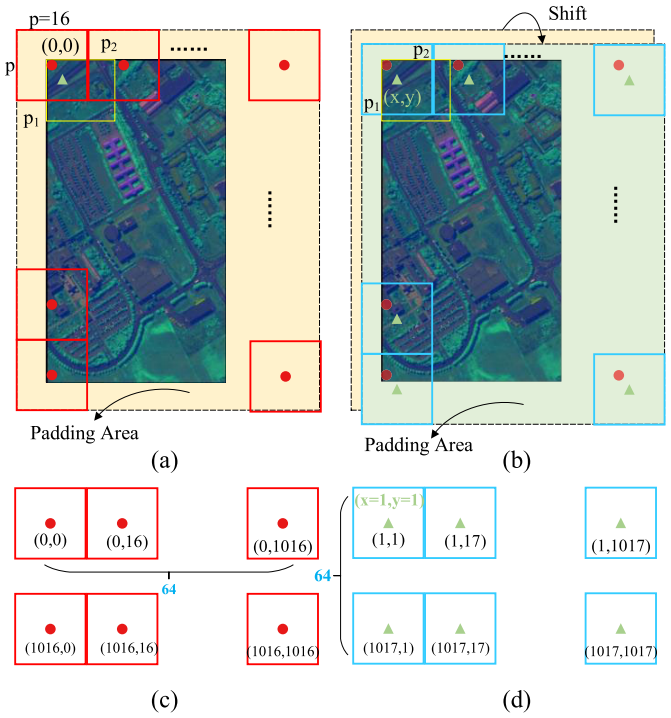


Fig. 2. Padded hyperspectral image with 64 × 64 pixels' feature under different padding values. The red and blue frames are padded patches. The yellow frames are patches divided in hyperspectral data. (a) and (b) Represent the padded hyperspectral images with padding values p1 and p2 set to (8,8) and (7,7), respectively. (c) and (d) Show the coordinates of the 64 × 64 pixel features extracted from (a) and (b), respectively.

the padding size can be expressed as follows:

$$x = \frac{3}{2}p - p_1, \quad \frac{p}{2} + 1 \le p_1 \le \frac{p}{2} + p \tag{1}$$

$$y = \frac{3}{2}p - p_2, \quad \frac{p}{2} + 1 \le p_2 \le \frac{p}{2} + p \tag{2}$$

where p_1 and p_2 are the padding values on the left and upper sides of the hyperspectral data. For a specific filling, we will get 64 × 64 features of pixels, as shown in Fig. 2(c) and (d). To obtain all features in the patches of hyperspectral data, we adjust the padding values to make a shift on the coordinates of obtained features in the patch of hyperspectral data, in which the number of shifts corresponds to the total number of samples, namely, $p \times p$ in the hyperspectral data patches. In different patches, when the padding values p_1 and p_2 are equal to $(p/2) + p + 1$, the features of a total of 64 pixels with coordinates between (0, 0) and (1016, 1016) are obtained, as shown in Fig. 2(c).

Through the self-adaptive padding mechanism, the dimension of the reduced hyperspectral data $\acute{X}$ is first padded to 1024 × 1024 through mirror completion and then segmented into patches. Each patch P has a size of 16 × 16, resulting in a total of 64 × 64 patches. If one of these patches is denoted as P^*, the cropped data can be represented as follows:

$$\acute{X} = \left\{P_1, P_2, P_n^*, \ldots, P_{4096}\right\}. \tag{3}$$

For a single patch with a size of $p \times p$ in hyperspectral data, each padding can only obtain one pixel's feature in this patch.

Specifically, the processing steps of GFEsNet are given as follows. The flowchart of GFEsNet is shown in Fig. 3.

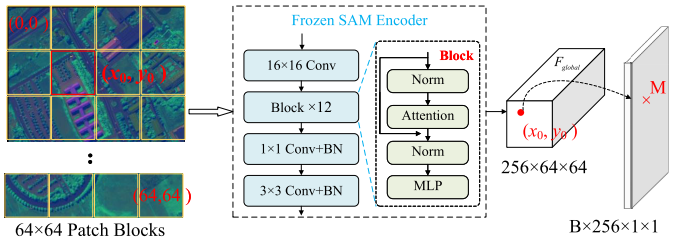


Fig. 3. Flowchart of the GFEsNet, which demonstrates the coordinate correspondence between the input and output features. The input padding is omitted.

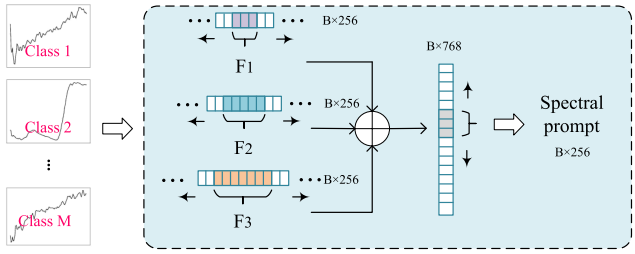


Fig. 4. Flowchart of the SFPsNet, where M denotes the number of categories, $\oplus$ signifies the feature concatenation, and B means the batch size.

When extracting features for P_n^* under single padding, all 64 × 64 patches with a size of 16 × 16 are input into the encoder with frozen parameters, which is shown in Fig. 3, and each patch possesses its own positional encoding information. The output feature F_{global}, processed by the encoder, has dimensions of 256 × 64 × 64. Each pixel in the feature F_{global} corresponds to the respective patch at its corresponding position. The F_{global} can be defined as

$$F_{\text{global}_i} = \text{Encoder}(P_i). \tag{4}$$

Finally, based on the coordinate information of the pixel, the preprocessed feature of that pixel from P_n^* is extracted and incorporated into the output feature $\hat{F}$ of GFEsNet. The position information is recorded for subsequent supervised training. This process is repeated for all training samples.

B. SFPsNet

In the SAGFFNet framework, a global feature struggles to capture all feature information. The motivation behind the design of SFPsNet which is described in Fig. 4 is to enhance the interpretability of the global features extracted by GFEsNet by learning the spectral feature information for different categories, enabling the model output to more comprehensively describe the spectral behavior of all categories. Given the importance of spectral texture information in HSIC, it is crucial for SFPsNet to retain spectral information at different scales, thereby enhancing the effectiveness of the spectral prompts.

Specifically, due to the proximity of spectral information between adjacent spectral channels in HSI and the ability of convolution operations to retain spectral information from neighboring channels [35], 1-D convolutions of varying sizes are applied to the input spectral sequence to effectively capture the spectral information F_i, with convolution kernel sizes of 3, 5, and 7. This process can be expressed by the

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following formulas:

$$F_1 = \text{Relu}[\text{BN}(\text{Conv}_3(x_i))] \tag{5}$$

$$F_2 = \text{Relu}[\text{BN}(\text{Conv}_5(x_i))] \tag{6}$$

$$F_3 = \text{Relu}[\text{BN}(\text{Conv}_7(x_i))] \tag{7}$$

where x_i represents the 1-D spectral input for different samples, BN denotes the batch normalization, and Relu refers to the rectified linear unit.

To reduce computational complexity, the above three feature maps are aggregated along the spectral dimension and then apply a 1-D convolution with a kernel size of 3. The feature concatenation is denoted by Concat; thus, the process can be described as

$$F_{\text{local}} = \text{Conv}_3(\text{Concat}(F_1, F_2, F_3)). \tag{8}$$

The final output, which contains multiscale spectral information, is fused with the global features using a weighted summation to compensate for the algorithm's deficiency in spectral discrimination.

C. SIDsNet

SIDsNet is introduced as a semantic information decoder. Its primary goal is to integrate global spatial–spectral features with spectral prompt features extracted by SFPsNet, aiming to improve the overall classification performance. By efficiently fusing these features, SIDsNet facilitates a more comprehensive understanding and representation of hyperspectral data, thereby advancing classification accuracy.

A joint spectral–spatial weighted (JSSW) fusion is developed to fuse global features with spectral prompts. Specifically, by integrating global features and spectral prompts, we designed two learnable weights α and β, which are applied to global features and spectral prompts, respectively, to balance the importance of global features and spectral prompts. This approach not only simplifies the network architecture but also enhances the computational efficiency. This process can be described as follows:

$$\alpha, \beta = \text{FC}(\text{Concat}(F_{\text{global}}, F_{\text{local}})) \tag{9}$$

$$F = \text{Concat}(\alpha \times F_{\text{global}}, \beta \times F_{\text{local}}). \tag{10}$$

To ensure that the fused features remain on a unified scale, we introduce constraints within the dynamic fusion process to further enhance computational stability. The constraint can be formulated as

$$\alpha + \beta = 1. \tag{11}$$

Finally, the aforementioned features are fed into the classifier to achieve the final classification probability output. To simplify the network complexity, the classifier is implemented using a single convolution layer. The process of convolution is expressed as

$$\hat{\mathbf{F}} = \sigma(\boldsymbol{W} \cdot BN(\mathbf{F}) + \boldsymbol{b}) \tag{12}$$

where $\text{BN}(\cdot)$ and $\sigma(\cdot)$ denote the feature normalization and the activation function, $\hat{\mathbf{F}}$ is the output of the activation function, and $\boldsymbol{W}$ and $\boldsymbol{b}$ denote the weight of the convolution kernel and bias, respectively.

The training process is supervised by cross-entropy loss $\mathscr{L}$, shown as follows:

$$\mathscr{L} = -\sum_{m=1}^{M} \mathbf{O}_m(y_i) \log\left(\mathbf{Y}_m^p(\boldsymbol{x}_i)\right) \tag{13}$$

where $\mathbf{O}_m(y_i)$ denotes the one-hot encoding format of the label y_i, $\mathbf{Y}_m^p$ is the predicated probability of sample $\boldsymbol{x}_i$ that belongs to the mth category, and M is the number of categories. The complete process of the SAGFFNet is shown in Algorithm 1.

Algorithm 1 Proposed Method

Training Phase:
Input: training samples: $D_{train} : \{x_i, y_i\}_{i=1}^n$, $Episodes$
Output: Updated parameters φ

```
Dimensionality reduction: x_i' = PCA(x_i)
for e in Episodes do
    Extract feature from training samples: F_global = f_φ^GFEs(x_i') and F_local = f_φ^SFPs(x_i)
    Get predict label: y_i' = f_φ^SIDs(F_global + F_local)
    Calculate L by (13)
    Update f_φ by optimizing L
EndFor
return updated parameters φ
```

Testing Phase:
Input: testing samples: $D_{test} : \{x_j, y_j\}_{j=1}^n$
Output: Classification results Y_{result}

```
Dimensionality reduction: x_j' = PCA(x_j)
Extract feature from training samples: F_global = f_φ^GFEs(x_j') and F_local = f_φ^SFPs(x_j)
Get predict label: y_j' = f_φ^SIDs(F_global + F_local)
return classification results: Y_result = {y_j'}_{j=1}^n
```

III. Experiments

In this section, four widely used datasets and the configurations of the experimental parameters are initially described. To validate the efficacy of the proposed algorithm, the performance of our proposed algorithm is evaluated against nine benchmark algorithms. The efficacy of the network's structure is analyzed through ablation studies subsequently. Then, the effectiveness of the network under various training sample ratios, conditions of limited sample availability, scenarios involving image degradation, and dimensionality reduction methods are explored.

A. Data Description

In this study, we utilize four widely recognized hyperspectral datasets to evaluate the performance of our proposed algorithm. These datasets, which vary in their spatial and spectral characteristics, offer a comprehensive testing ground for validating the algorithm's classification capabilities under different conditions. The datasets include the University of Pavia (UP) [36], Salinas (SA) [37], Pavia Center (PC) [38],

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and Houston2013 (HU) [18] datasets. Detailed descriptions of each dataset are provided in the following.

The UP dataset has 103 spectral bands with dimensions of 610 × 340 pixels and a spatial resolution of 1.3 m. The spectral range is from 0.43 to 0.86 μm, including reference data for nine classes. This dataset, characterized by complex spatial textures and irregular shapes of ground objects, is suitable for validating the classification performance of algorithms.

The SA dataset comprises 204 spectral bands with dimensions of 512 × 217 pixels and a spatial resolution of approximately 3.7 m. It includes a large number of classes with similar ground object categories, making it suitable for validating the algorithm's ability to distinguish between similar objects.

The PC dataset features an image with dimensions of 1096 × 715 pixels, a geometric resolution of 1.3 m, and 102 spectral bands. The ground truth identifies nine classes. This dataset is similar to the UP dataset, both commonly used for classification tasks.

The HU dataset contains 144 spectral bands with dimensions of 349 × 1905 pixels and a spatial resolution of 2.5 m, spanning a spectral range of 0.38–1.05 μm, with ground-truth images including 15 classes. This dataset features complex ground objects with numerous overlapping and variably sized objects, and some areas obscured by clouds, resulting in significant spectral variability. These characteristics make it particularly suitable for validating the classification performance of algorithms.

B. Experimental Configuration

The effectiveness of the algorithm is quantitatively assessed through overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) [39]. In addition, the training time (Train) and the testing time (Test) required when executed on the UP dataset are used to evaluate performance.

In our experiments, SAGFFNet is compared with nine DL methods to analyze algorithm effectiveness comprehensively. The compared methods are the 2DCNN [40], morFormer [41], SFFTT [42], SSTN [43], GTFN [44], TriCNN [45], ASPN [46], GSCVIT [47], and DBCTNet [48].

The SAGFFNet algorithm is implemented in Python using the PyTorch framework, with an initial learning rate set at 0.001, employing the cross-entropy loss function and the Adam optimizer. The SAGFFNet algorithm undergoes 128 training epochs. The primary hardware specifications for all algorithm implementations include: 1) CPU: Xeon Gold 8163; 2) GPU: GTX-4090; and 3) RAM memory: 128 GB. The input patch size is set to 16 × 16. Note that for each category of UP, SA, PC, and HU datasets, 1%, 1%, and 98% of the samples were randomly selected as training, validating, and testing datasets, respectively. The number of training and testing samples for each dataset is shown in Tables I–IV.

To ensure a fair comparison, the number of training samples for each algorithm is standardized. The experimental data presented in this section are derived from ten parallel repetitions of the experiment.

TABLE I
CATEGORY INFORMATION OF THE UP DATASET

Class ID	Class Name	Train.	Test.
1	Asphalt	67	6564
2	Meadows	187	18462
3	Gravel	21	2078
4	Trees/Grass	31	3033
5	Metalsheets	14	1331
6	Baresoil	51	4978
7	Bitumen	14	1316
8	Bricks	37	3645
9	Shadows	10	937
	Total	432	42344

TABLE II
CATEGORY INFORMATION OF THE SA DATASET

Class ID	Class Name	Train.	Test.
1	Brocoli_green_weeds_1	21	1988
2	Brocoli_green_weeds_2	38	3688
3	Fallow	20	1956
4	Fallow_rough_plow	14	1380
5	Fallow_smooth	27	2651
6	Stubble	40	3919
7	Celery	36	3543
8	Grapes_untrained	113	11158
9	Soil_vinyard_develop	63	6140
10	Corn_senesced_green_weeds	33	3245
11	Lettuce_romaine_4wk	11	1057
12	Lettuce_romaine_5wk	20	1907
13	Lettuce_romaine_6wk	10	906
14	Lettuce_romaine_7wk	11	1059
15	Vinyard_untrained	73	7195
16	Vinyard_vertical_trellis	19	1788
	Total	549	53580

TABLE III
CATEGORY INFORMATION OF THE PC DATASET

Class ID	Class Name	Train.	Test.
1	Water	660	65311
2	Trees	76	7522
3	Asphalt	31	3059
4	Self-Blocking Bricks	27	2658
5	Bitumen	66	6518
6	Tiles	93	9155
7	Shadows	73	7214
8	Meadows	429	42397
9	Bare Soil	29	2834
	Total	1484	146668

C. Analysis of Classification Results

This section compares and analyzes the classification performance of all algorithms. Figs. 5–8 display the classification results of all methods on the UP, SA, PC, and HU datasets. Tables V–VIII illustrate the overall classification accuracy and each category's accuracy of the above ten comparative algorithms on the UP, SA, PC, and HU datasets.

In this section, we used the analysis of variance (ANOVA) [49] to statistically analyze the significance of differences in classification performance (OA) between the proposed algorithm and other comparative algorithms. Methods that exhibit a statistically significant performance difference compared to our algorithm are marked with an asterisk (*) in the OA rows of Tables V–VIII.

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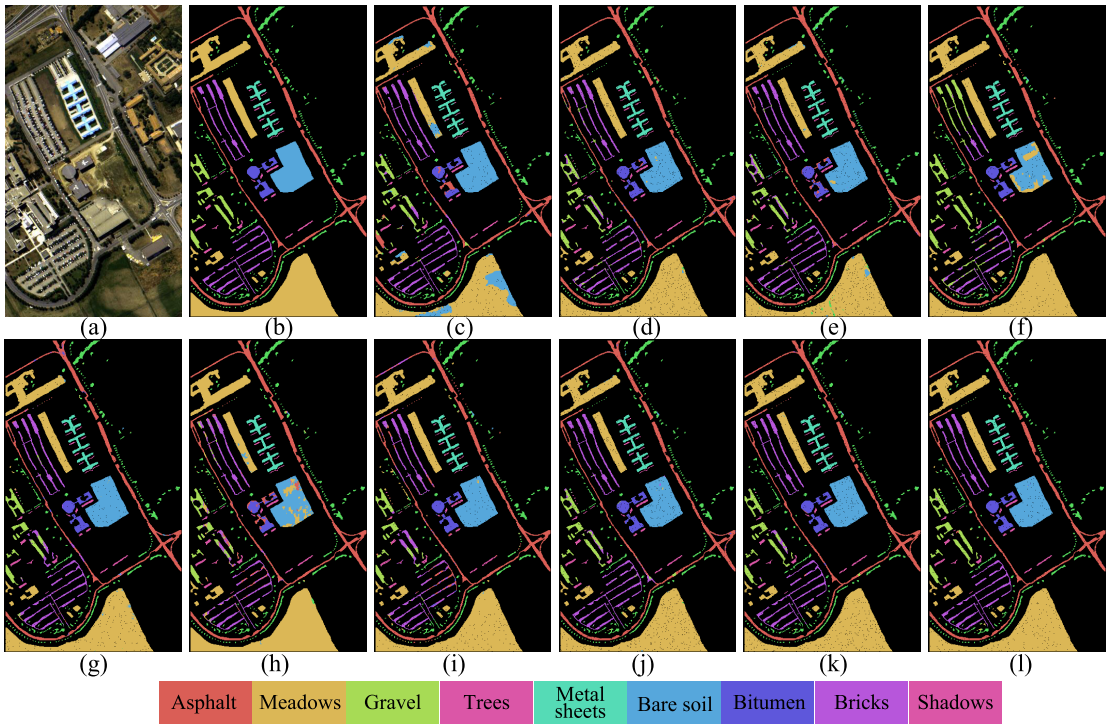


Fig. 5. Classification maps of all different methods on the UP dataset. (a) False-color image. (b) Ground truth. (c) 2DCNN. (d) morFormer. (e) SSFTT. (f) SSTN. (g) GTFN. (h) TriCNN. (i) ASPN. (j) GSCVIT. (k) DBCTNet. (l) SAGFFNet.

TABLE IV
CATEGORY INFORMATION OF THE HU DATASET

Class ID	Class Name	Train.	Test.
1	Healthy grass	15	1359
2	Stressed grass	14	1440
3	Synthetic grass	9	786
4	Trees	13	1251
5	Soil	13	1285
6	Water	3	336
7	Residential	16	1460
8	Commercial	15	1339
9	Road	17	1537
10	Highway	16	1408
11	Railway	18	1548
12	Parking Lot 1	14	1415
13	Parking Lot 2	6	626
14	Tennis Court	5	508
15	Running Track	8	790
	Total	182	17088

The proposed algorithm exhibits consistently exceptional classification performance across all four datasets—UP, SA, PC, and HU—demonstrating its robustness and effectiveness in various contexts. On the UP dataset, the algorithm achieves an overall accuracy of 99.32%, outshining all other methods. This high performance is corroborated by the highest average accuracy (99.07%) and Kappa coefficient (99.10%). The results indicate the algorithm's reliable performance in distinguishing different classes. Similarly, the algorithm excels on the SA dataset with an overall accuracy of 99.61%, the highest among all compared methods. This performance highlights the algorithm's ability to handle complex and varied data distributions effectively, providing consistent and accurate classification results. On the PC dataset, the proposed algorithm again stands out, achieving an overall accuracy of 99.48%, which is superior to all other algorithms. The average accuracy is 98.38%, and the Kappa coefficient is the highest at 99.27%. These metrics underscore the algorithm's robustness in handling high-dimensional data and its reliability in maintaining high classification performance across different evaluation criteria. On the HU dataset, the algorithm achieves an overall accuracy of 91.48%, while lowers than in other datasets. Despite the challenging nature of the HU dataset, the algorithm maintains competitive performance, demonstrating its adaptability to different data characteristics and its ability to deliver reliable results even in less favorable conditions. In summary, the proposed algorithm demonstrates exceptional performance across a diverse set of datasets. This consistent superiority across various metrics and datasets highlights the algorithm's effectiveness and robustness in classification tasks, making it a valuable tool for high-accuracy classification in diverse scenarios.

More specifically, CNN-based algorithms such as 2DCNN and TriCNN demonstrate strong classification performance across all datasets. However, compared to transformer framework-based methods such as SSFTT, GSCVIT, GTFN, and morFormer, CNN-based methods show a disparity in classification accuracy. The transformer framework's ability to handle sequence features effectively allows it to exploit spectral characteristics in HSI data and incorporate global attention mechanisms. This capability enhances model performance across various spatial and spectral resolutions by capturing interband relationships, thereby improving accuracy in HSIC and target detection.

In contrast to transformer-based methods, the SAGFFNet algorithm, which is based on a foundation model, maintains the best classification results on all four datasets.

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TABLE V
CLASSIFICATION ACCURACY (%) OF ALL ALGORITHMS ON THE UP DATASET WITH 1% TRAINING SAMPLE

Type	2DCNN	morFormer	SSFTT	SSTN	GTFN	TriCNN	ASPN	GSCVIT	DBCTNet	**SAGFFNet**
1	95.56	97.57	95.69	97.21	97.72	96.03	89.48	98.01	97.94	**99.00**
2	98.62	99.39	99.46	94.31	99.56	99.56	98.61	98.88	99.10	**99.77**
3	84.80	81.28	95.05	90.12	81.43	83.15	87.01	89.61	93.01	**96.57**
4	94.51	96.84	96.75	97.19	94.03	93.43	97.51	94.60	96.38	**99.19**
5	97.21	99.83	99.55	99.26	99.21	**99.96**	97.39	99.83	99.77	99.62
6	96.25	99.28	**99.74**	99.28	98.45	89.30	99.01	97.72	99.12	99.46
7	85.05	90.68	92.68	95.03	97.99	51.93	96.94	94.34	98.84	**99.13**
8	93.39	93.53	89.17	86.73	93.62	73.19	78.26	95.39	97.46	**98.88**
9	90.94	98.53	92.71	98.08	96.43	99.25	92.67	97.91	97.84	**99.98**
OA	95.92*	97.25	97.24	94.23*	97.22	92.82*	94.52*	97.41	98.27	**99.32**
AA	91.41	95.22	94.99	95.25	95.38	87.31	92.99	96.25	97.72	**99.07**
Kappa	94.56	96.35	96.34	92.25	96.31	90.40	92.70	96.57	97.71	**99.10**

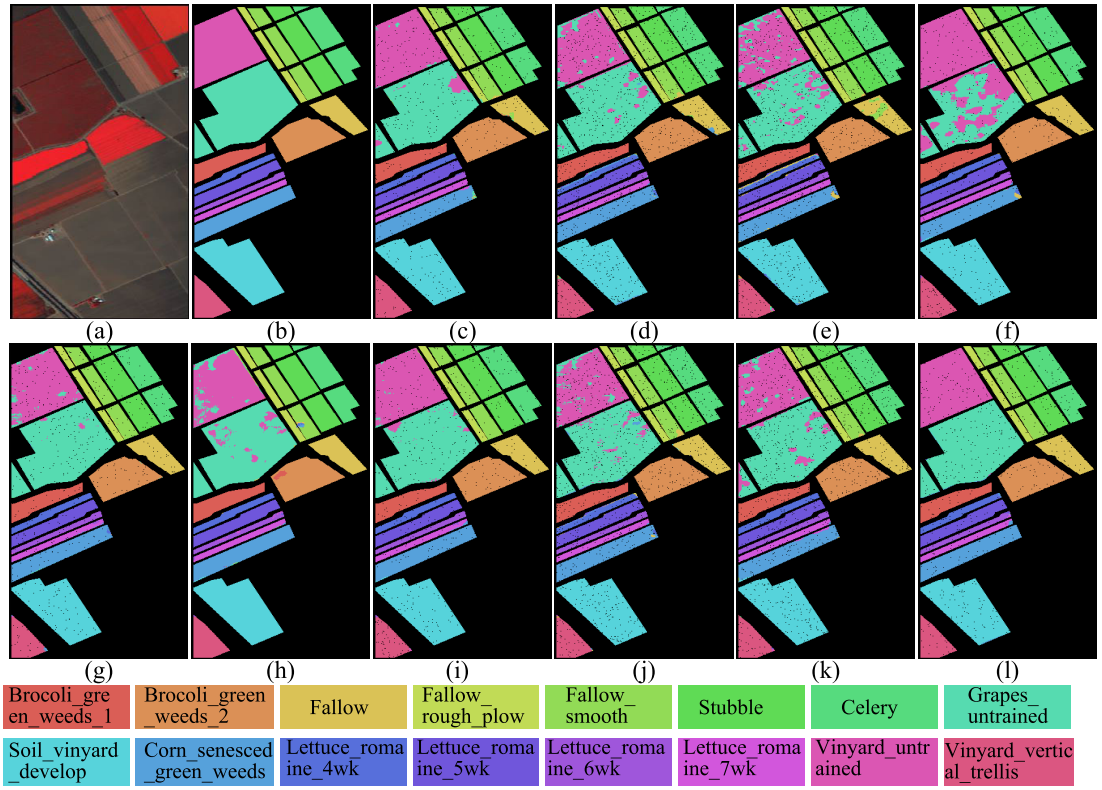


Fig. 6. Classification maps of all different methods on the SA dataset. (a) False-color image. (b) Ground truth. (c) 2DCNN. (d) morFormer. (e) SSFTT. (f) SSTN. (g) GTFN. (h) TriCNN. (i) ASPN. (j) GSCVIT. (k) DBCTNet. (l) SAGFFNet.

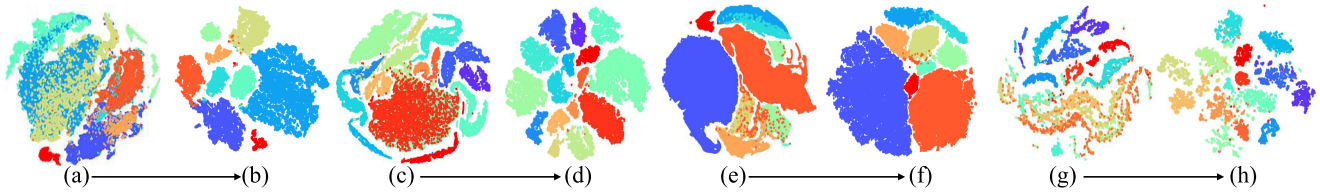


Fig. 7. Feature distribution (t-SNE visualization) of four datasets. (a) UP. (b) UP processed by SAGFFNet. (c) SA. (d) SA processed by SAGFFNet. (e) PC. (f) PC processed by SAGFFNet. (g) HU. (h) HU processed by SAGFFNet.

Specifically, the proposed algorithm integrates a pretrained SAM encoder, leveraging its powerful information perception and feature extraction capabilities. For each patch, the GFEsNet of the SAGFFNet algorithm effectively integrates information from all patches, aiding in mitigating spectral variations and extracting more accurate global spatial–spectral information, thus fully tapping into the potential of foundation models in the field of HSIC. The SFPsNet, focusing on the deficiency of transformer-based framework in capturing local spectral information, effectively fuses local spectral features, thereby enhancing the overall classification accuracy.

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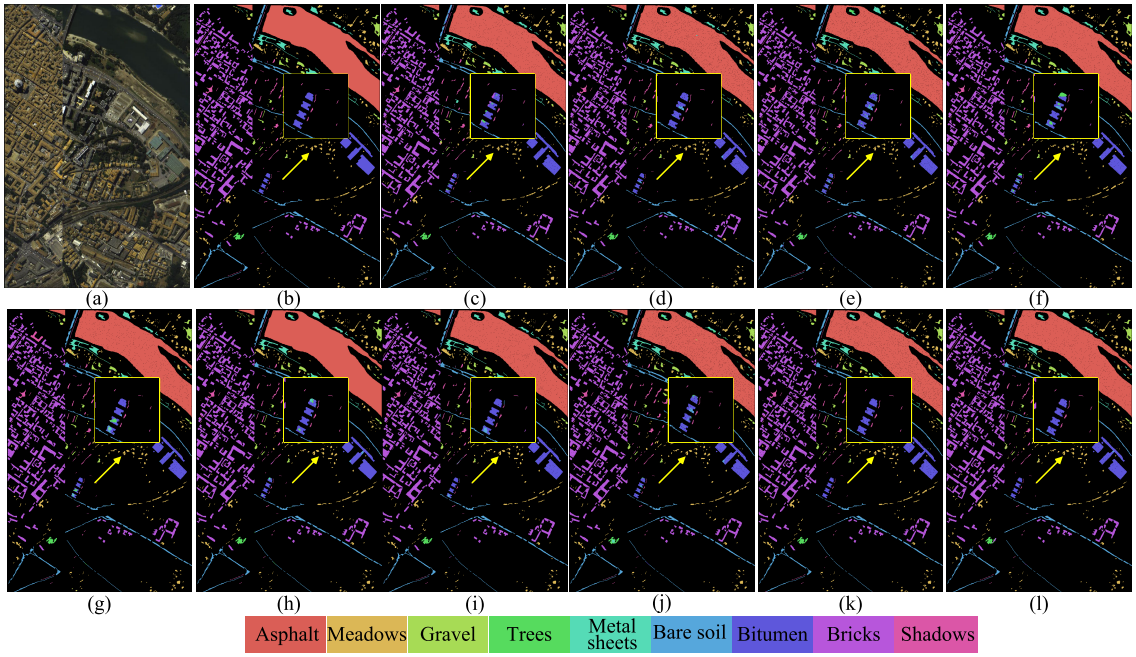


Fig. 8. Classification maps of all different methods on the PC dataset. (a) False-color image. (b) Ground truth. (c) 2DCNN. (d) morFormer. (e) SSFTT. (f) SSTN. (g) GTFN. (h) TriCNN. (i) ASPN. (j) GSCVIT. (k) DBCTNet. (l) SAGFFNet.

TABLE VI

CLASSIFICATION ACCURACY (%) OF ALL ALGORITHMS ON THE SA DATASET WITH 1% TRAINING SAMPLE

Type	2DCNN	morFormer	SSFTT	SSTN	GTFN	TriCNN	ASPN	GSCVIT	DBCTNet	**SAGFFNet**
1	99.23	97.56	99.99	95.35	99.98	**100**	**100**	98.31	99.76	99.61
2	99.59	99.96	**100**	97.62	99.95	**100**	99.60	99.95	98.88	**99.99**
3	98.56	94.05	**100**	95.99	99.98	99.78	99.95	97.01	99.80	99.93
4	98.51	98.42	97.60	98.62	99.52	98.39	96.61	97.51	99.62	**99.81**
5	98.65	94.90	98.87	**99.06**	96.95	96.42	98.14	97.67	98.68	**99.06**
6	99.87	**100**	**100**	99.95	99.91	**100**	99.88	99.99	99.95	99.84
7	99.68	99.78	**99.99**	98.17	99.90	99.90	99.97	99.77	99.90	99.72
8	92.67	91.09	98.22	93.49	97.61	90.00	97.31	93.26	77.00	**99.27**
9	99.38	99.54	99.92	99.49	**99.99**	99.90	99.71	99.80	99.91	99.59
10	98.72	96.68	**99.69**	97.83	97.46	97.46	98.93	98.29	95.52	98.42
11	97.61	93.09	**100**	95.97	99.72	99.64	99.93	96.87	99.33	97.56
12	99.30	98.90	99.95	98.33	99.23	95.03	99.96	98.44	99.84	**100**
13	97.31	**99.76**	99.33	99.17	99.14	69.92	98.14	98.73	98.98	98.12
14	97.54	97.56	97.95	98.86	93.53	96.50	97.42	**99.29**	98.92	95.10
15	88.32	90.53	98.32	83.77	97.06	85.33	95.53	89.18	84.37	**99.94**
16	99.14	98.88	99.99	99.39	99.47	99.45	**100**	98.84	99.31	99.42
OA	96.41*	95.74*	99.20	94.73*	98.58	94.77*	98.37	96.46*	92.58*	**99.61**
AA	97.69	96.92	99.40	96.94	98.71	95.48	98.77	97.68	96.86	**99.41**
Kappa	96.00	95.25	99.11	94.14	98.42	94.17	98.19	96.06	91.78	**99.56**

TABLE VII

CLASSIFICATION ACCURACY (%) OF ALL ALGORITHMS ON THE PC DATASET WITH 1% TRAINING SAMPLE

Type	2DCNN	morFormer	SSFTT	SSTN	GTFN	TriCNN	ASPN	GSCVIT	DBCTNet	**SAGFFNet**
1	**100**	**100**	99.98	99.98	**100**	**100**	99.99	99.28	99.66	**100**
2	97.32	97.30	**99.01**	96.40	97.90	93.59	98.03	97.96	95.50	97.43
3	91.70	92.26	89.97	92.67	96.18	94.92	92.08	89.62	**96.58**	93.76
4	97.54	97.31	99.66	94.12	96.22	89.78	98.76	96.34	**99.71**	97.15
5	98.55	99.39	**99.92**	96.64	99.11	96.93	97.45	97.46	98.96	99.66
6	98.95	99.20	97.42	98.47	99.20	99.10	96.18	99.27	**99.58**	99.48
7	99.60	97.79	**99.99**	98.89	96.76	94.27	99.81	97.36	96.77	99.28
8	99.17	99.91	99.62	99.79	99.45	99.75	99.63	**99.93**	99.32	99.75
9	95.07	99.03	98.04	98.54	97.55	95.44	97.76	99.05	99.18	**99.99**
OA	99.26*	99.42	99.39	99.13*	99.29	98.74*	99.19*	99.28	99.10*	**99.48**
AA	97.32	98.02	98.15	97.28	98.04	95.98	97.74	97.44	98.36	**98.38**
Kappa	98.96	99.18	99.14	98.77	98.99	98.22	98.85	98.99	98.73	**99.27**

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TABLE VIII
CLASSIFICATION ACCURACY (%) OF ALL ALGORITHMS ON THE HU DATASET WITH 1% TRAINING SAMPLE

Type	2DCNN	morFormer	SSFTT	SSTN	GTFN	TriCNN	ASPN	GSCVIT	DBCTNet	**SAGFFNet**
1	83.20	93.10	82.29	86.14	88.60	84.33	91.87	90.93	90.61	**91.38**
2	82.14	88.58	91.72	87.07	88.79	91.97	88.51	92.31	87.85	**94.53**
3	89.24	96.17	**99.99**	95.29	96.75	95.34	99.73	97.05	98.96	99.95
4	87.33	94.64	**94.98**	92.63	90.85	87.84	90.73	94.52	94.51	91.38
5	90.89	97.42	94.43	92.64	97.67	**99.98**	98.01	94.79	98.97	99.03
6	59.93	86.08	97.18	97.38	82.02	88.03	**98.81**	83.87	89.81	77.12
7	83.77	84.26	92.66	79.56	77.72	50.19	83.69	78.43	89.53	**94.31**
8	66.53	72.34	**99.92**	84.44	60.95	63.38	95.83	63.82	59.33	66.51
9	78.82	84.77	87.50	81.68	71.03	46.48	76.85	76.81	73.05	**93.51**
10	80.05	77.56	69.27	65.54	88.07	75.75	74.05	68.45	46.43	**97.71**
11	88.27	70.57	76.14	76.61	92.20	60.72	87.56	74.19	60.69	**97.75**
12	77.78	79.17	81.83	66.64	67.89	79.19	77.64	72.40	50.59	**89.48**
13	81.22	89.89	83.00	69.10	83.35	80.51	84.56	80.86	**94.36**	75.34
14	90.11	91.85	**100**	97.28	98.86	98.70	**100**	89.78	98.84	94.74
15	86.90	98.52	90.93	93.21	98.46	**99.94**	95.73	98.90	99.91	98.58
OA	82.41*	85.65*	86.35*	81.45*	84.17*	76.76*	86.88*	82.34*	78.73*	**91.48**
AA	80.46	86.99	87.27	84.36	85.55	80.16	89.57	83.81	82.28	**90.96**
Kappa	80.96	84.49	85.25	79.93	82.90	74.90	85.82	80.91	77.05	**91.18**

TABLE IX
OA (%) OF THE ABLATION EXPERIMENTS FOR DIFFERENT PARTS OF THE SAGFFNET ALGORITHM

Type	UP	SA	PC	HU
remove SFPsNet	98.62±0.19	99.30±0.18	99.21±0.14	80.04±0.94
remove SIDsNet	99.01±0.14	98.53±0.13	99.29±0.07	90.20±0.50
remove GFEsNet	95.67±0.54	96.32±0.38	99.00±0.09	83.37±1.41
SAGFFNet	99.32±0.08	99.44±0.12	99.48±0.09	91.84±0.23

TABLE X
OA (%) OF THE SAGFFNET FOR DIFFERENT FUSION TECHNIQUES

Method	UP	SA	PC	HU
Attention	98.45±0.24	98.92±0.13	99.12±0.12	89.51±0.55
Concat	99.08±0.21	99.18±0.24	99.21±0.18	89.84±0.69
JSSW	**99.32±0.08**	**99.44±0.12**	**99.48±0.09**	**91.84±0.23**

The feature distributions before and after SAGFFNet processing are shown in Fig. 7. It is evident that SAGFFNet effectively utilizes spatial–spectral features, which contributes to improved classification performance.

D. Influence of Different Model Components

To further explore the effectiveness of the proposed network structure, we conducted ablation experiments on four datasets to assess the impact of three modules of SAGFFNet on classification performance. The experimental results are summarized in Table IX, and the classification maps of the ablation experiments for different parts of the SAGFFNet algorithm are shown in Fig. 9.

The analysis from Fig. 9 reveals that removing GFEsNet has the most pronounced effect on algorithmic performance, causing the image classification boundaries to become blurred. Conversely, integrating GFEsNet clarifies object boundaries in classification results. The SFPsNet leads to enhanced classification accuracy, such as a 0.70% improvement on the UP dataset, indicating its role in bolstering SAGFFNet's capability to extract deep features using the foundation model. Furthermore, the inclusion of SIDsNet enhances model robustness, maximizing the foundation model's potential in HSIC.

E. Influence of Different Fusion Strategies

To investigate the effectiveness of different fusion strategies in SIDsNet for integrating global features and spectral prompts, we compared two fusion approaches: concat fusion and attention fusion. The concat fusion method simply combines the two types of features by concatenating them along the spectral channel, while the attention fusion method utilizes an attention mechanism to learn the weights of global features and spectral prompts separately. The classification results for each fusion strategy are shown in Table X. It can be seen that JSSW fusion has a better performance than other twos. JSSW fusion achieves dynamic integration by assigning constrained weights to both global features and spectral prompts, which facilitates a more robust fusion compared to the simple feature concatenation of concat fusion. Furthermore, it more effectively captures the intrinsic relationships between global features and spectral prompts than attention fusion, thus enhancing the model's classification capability.

F. Influence of Different Padding Sizes

In the self-adaptive padding mechanism, the padding size determines the ability of individual pixel samples to perceive surrounding spatial information. To explore the impact of different padding sizes on the algorithm's classification performance, we set the padding sizes to 5, 7, 9, 11, 13, and 15, respectively, with the results shown in Fig. 11. It is evident that as the padding size increases, the algorithm is able to capture more spatial information, and the classification performance shows an upward trend across the four datasets. The results on the HU dataset indicate that for spatially inconsistent distributions, the padding size can be reasonably adjusted to achieve optimal classification performance.

G. Influence of the Imaging Degradation

Hyperspectral data are sensitive to imaging degradation, which can significantly reduce algorithmic classification

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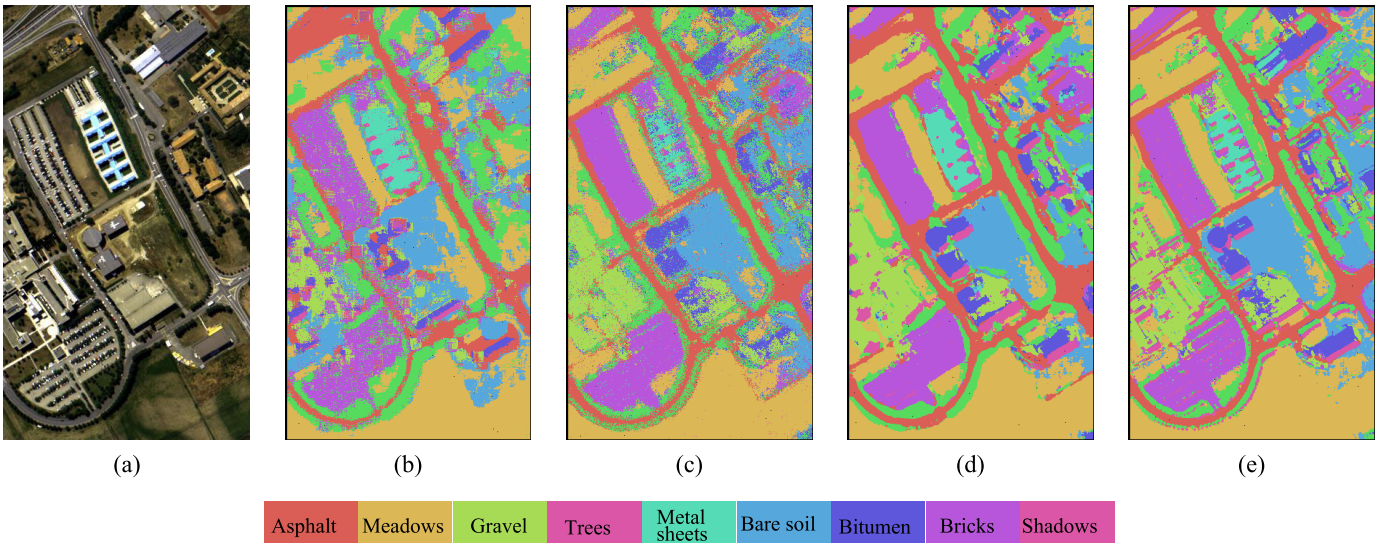


Fig. 9. Classification maps of the ablation experiments for different parts of the SAGFFNet algorithm on the UP dataset. (a) False-color image. (b) Remove GFEsNet. (c) Remove SIDsNet. (d) Remove SFPsNet. (e) SAGFFNet.

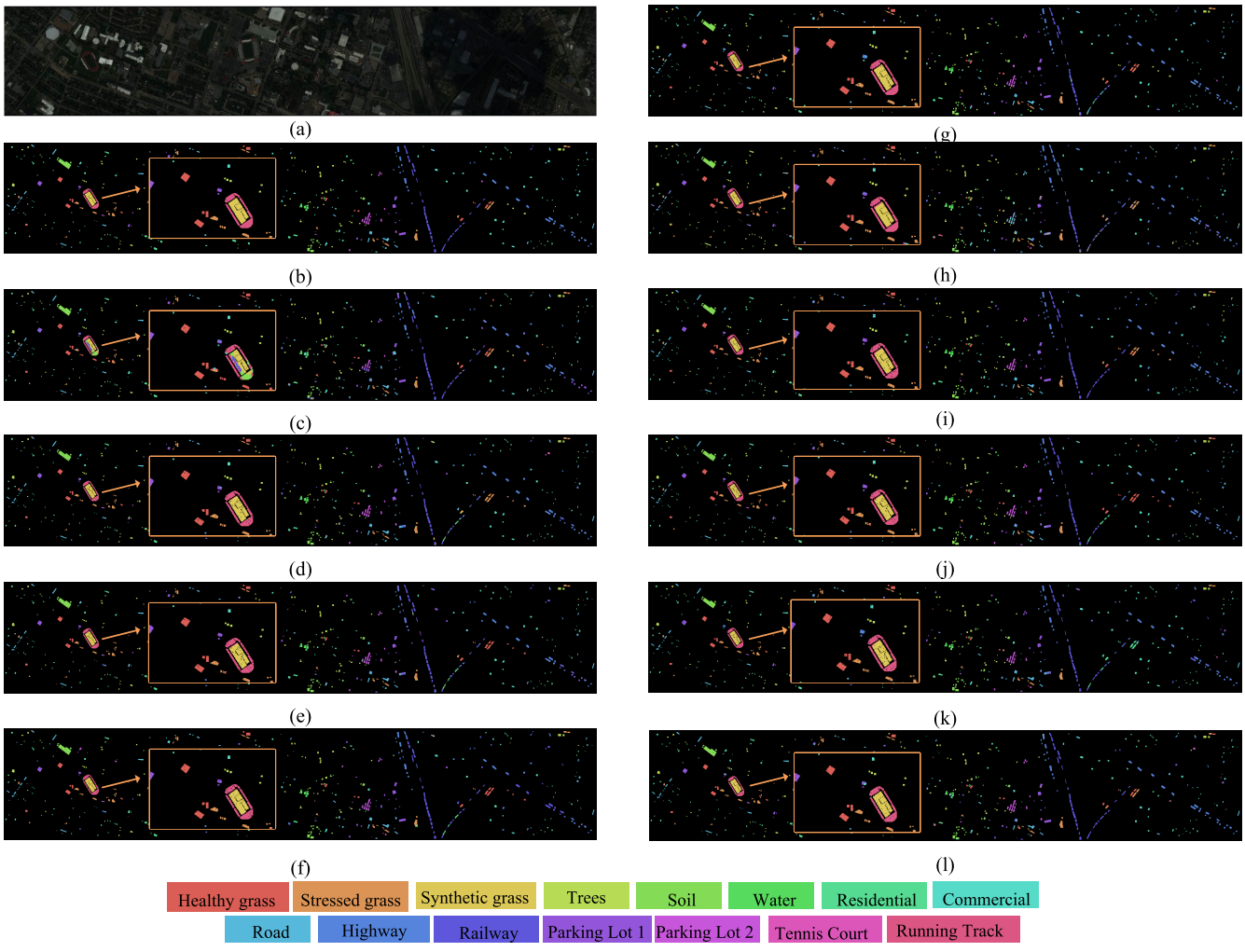


Fig. 10. Classification maps of all different methods on the HU dataset. (a) False-color image. (b) Ground truth. (c) 2DCNN. (d) morFormer. (e) SSFTT. (f) SSTN. (g) GTFN. (h) TriCNN. (i) ASPN. (j) GSCVIT. (k) DBCTNet. (l) SAGFFNet.

accuracy. In this study, we investigated the impact of imaging degradation on the proposed SAGFFNet algorithm. Following the methods outlined in reference [50] for introducing haze into hyperspectral datasets, we simulated haze on the UP

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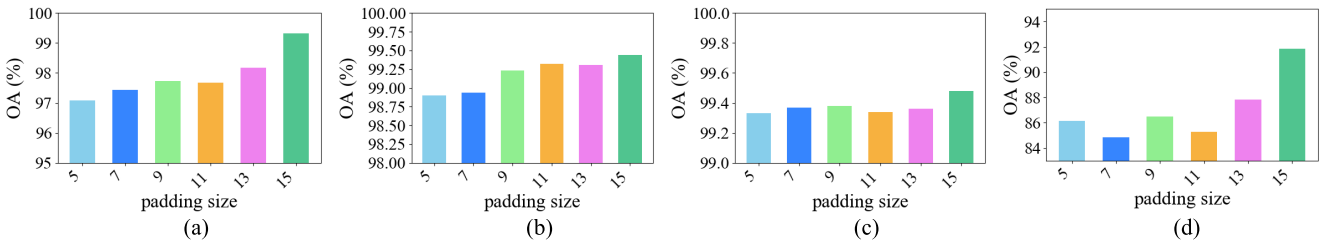


Fig. 11. OA (%) of SAGFFNet with different padding sizes. (a) UP. (b) SA. (c) PC. (d) HU.

TABLE XI
OA (%) OF ALL ALGORITHMS ON UP AND SIMULATED FOGGING DATASETS WITH 1% TRAINING SAMPLE

Type	2DCNN	morFormer	SSFTT	SSTN	GTFN	TriCNN	ASPN	GSCVIT	DBCTNet	**SAGFFNet**
No haze	95.92±1.35	97.25±0.58	97.24±0.34	94.23±0.58	97.22±0.06	92.82±0.81	94.52±0.25	97.41±0.53	98.27±0.52	99.53±0.12
Add haze	84.50±6.62	93.32±1.07	91.25±1.33	74.24±4.23	97.20±0.21	91.99±0.93	94.21±0.59	92.72±1.65	96.81±0.85	97.78±0.32

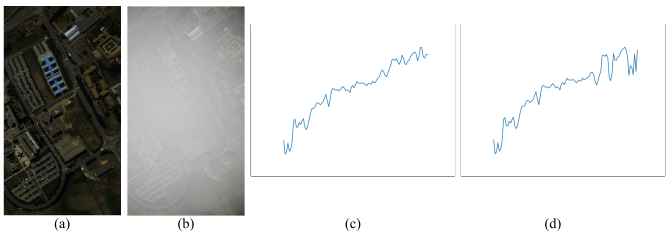


Fig. 12. UP and simulated datasets with their spectral curve. (a) UP. (b) UP with haze. (c) Spectral curve. (d) Spectral curve with haze.

TABLE XII
OA (%) OF THE SAGFFNET FOR VARIOUS DIMENSIONALITY REDUCTION TECHNIQUES

Method	UP	SA	PC	HU
SVD	99.03±0.21	99.34±0.09	99.31±0.12	90.99±1.04
NMF	99.07±0.23	**99.45±0.10**	99.44±0.06	89.85±0.67
PCA	**99.32±0.08**	99.44±0.12	**99.48±0.09**	**91.84±0.23**
MNF	98.27±0.26	97.98±0.24	99.26±0.06	85.18±1.66

dataset, which is shown in Fig. 12. The experiments were conducted with a training sample ratio of 1%. The results are presented in Table XI.

The experimental findings indicate that various forms of haze-induced imaging degradation led to varying degrees of performance decline across different methods. For instance, the SSFTT experienced a 5.99% decrease in classification accuracy following haze simulation. Notably, leveraging the deep feature extraction capabilities of its foundation model, SAGFFNet demonstrated superior classification accuracy compared to the aforementioned methods. Moreover, SAGFFNet effectively mitigates the impact of imaging degradation on classification performance.

H. Influence of Various Dimensionality Reduction Techniques

In our method, PCA is adopted as the dimensionality reduction method for hyperspectral data due to its simple implementation and easy-using. In practical applications, the methods for dimension reduction of hyperspectral data are not unique. In this section, we test the impact of three other dimensionality reduction methods, namely, singular value decomposition (SVD), nonnegative matrix factorization (NMF), and minimum noise fraction (MNF), on the classification performance of the proposed method. In our experiments, the number of principal components for all dimensionality reduction methods is set as 3. Other experimental parameters remain unchanged. The classification accuracies on four datasets under 1% training samples condition can be shown in Table XII.

As we can see, with different dimensionality reduction methods, our method still performs well, which proves its robustness and effectiveness. The classification performance shows a significant decline only on the HU dataset when using MNF, which is attributed to the noise reduction strategy that overly simplifies the feature information useful for classification. Besides, the dimensionality reduction methods that enable our method to achieve the best performance on different datasets are distinct, which indicates that various dimensionality reduction methods have different applicability on different HSIs. Considering that our algorithm exhibits strong robustness to the selection of various dimensionality reduction methods, in practice, we only need to choose an appropriate dimensionality reduction method according to experience.

I. Influence of the Number of Training Samples

To further investigate the influence of training sample ratios on the proposed algorithm, training sample ratios of 0.5%, 1%, 1.5%, 2%, and 2.5% were selected for the UP, SA, PC, and HU datasets, respectively, to evaluate the classification accuracy of the methods under consideration. The experimental results are illustrated in Fig. 13.

It is observed that as the training sample ratio increases across the four datasets, the classification accuracy of various methods generally improves. Fig. 13 shows that among all algorithms, SAGFFNet (depicted by the purple line) consistently achieves the highest classification accuracy across different training ratios. Notably, SAGFFNet demonstrates robust performance even with a minimal training sample ratio

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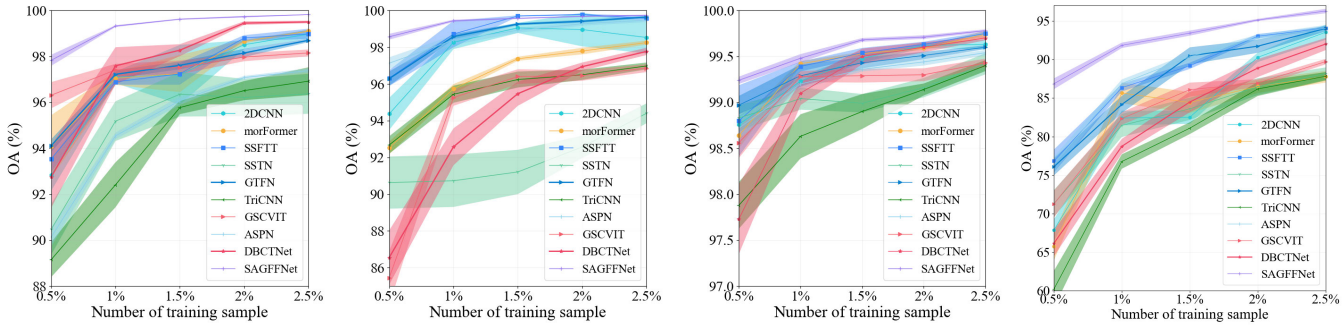


Fig. 13. Classification accuracies of SAGFFNet with the different number of training samples on the four datasets. (a) UP. (b) SA. (c) PC. (d) HU.

TABLE XIII
CLASSIFICATION ACCURACY (%) OF ALL ALGORITHMS ON FOUR DATASETS UNDER FEW SAMPLE CONDITIONS

Dataset	Num	2DCNN	morFormer	SSFTT	SSTN	GTFN	TriCNN	ASPN	GSCVIT	DBCTNet	**SAGFFNet**
PaviaU	5	62.26	77.51	75.72	63.54	68.26	62.41	63.12	78.12	76.71	**81.25**
	10	82.72	85.27	77.05	79.67	85.74	75.93	78.61	83.96	90.61	**91.04**
Salinas	5	77.84	79.18	83.90	82.54	88.48	82.46	**91.3**	88.49	86.66	88.80
	10	86.01	85.39	84.15	86.27	93.33	86.28	92.34	89.77	92.90	**96.50**
PaviaC	5	89.29	92.96	92.60	92.48	93.20	86.04	88.92	90.97	92.70	**95.49**
	10	96.05	95.23	96.89	97.05	96.65	90.92	94.94	96.80	96.79	**97.46**
Houston	5	74.82	73.81	74.06	67.21	74.13	60.85	73.39	73.50	74.34	**81.34**
	10	76.14	79.89	78.90	76.45	82.73	68.20	81.58	81.40	85.60	**87.16**

TABLE XIV
TRAINING, TESTING TIME (S), AND THE NUMBER OF TRAINED PARAMETERS (M) OF ALL METHODS ON FOUR DATASETS

	Type	2DCNN	morFormer	SSFTT	SSTN	GTFN	TriCNN	ASPN	GSCVIT	DBCTNet	**SAGFFNet**
UP	Train	45.03	68.51	24.74	31.51	142.28	74.81	14.87	47.75	68.64	99.77
	Test	4.41	11.72	8.09	4.99	23.85	6.04	1.95	8.04	4.52	3.41
	Params	0.1237	0.2268	0.4891	0.1562	0.1208	17.9570	0.1060	0.0927	0.0016	1.7409
SA	Train	99.69	68.53	40.42	38.78	177.28	68.43	16.98	54.72	122.25	122.15
	Test	8.05	15.84	5.98	6.27	29.50	6.57	1.18	9.31	6.01	4.61
	Params	0.1537	0.2807	0.9550	0.2069	0.1213	17.9572	0.6766	0.1490	0.0038	4.3934
PC	Train	160.46	250.58	109.71	106.69	504.54	148.71	15.4	162.11	270.71	348.09
	Test	14.12	20.02	12.79	16.47	79.85	17.38	2.95	14.85	15.05	12.18
	Params	0.1234	0.1986	0.4845	0.1557	0.1208	17.9570	0.1042	0.1265	0.0016	1.7231
HU	Train	23.39	104.93	30.79	12.12	57.93	31.16	12.67	63.52	34.71	55.58
	Test	4.17	14.68	12.25	2.02	9.61	2.16	1.15	8.20	2.97	1.29
	Params	0.1363	0.2326	0.6787	0.1177	0.1212	17.9572	0.3217	0.1035	0.0022	2.6137

of 0.5%, highlighting its effectiveness with limited training data. Compared with other methods, SAGFFNet effectively leverages a pretrained foundation model to extract global spectral features and integrates local spectral features through fusion mechanisms, thereby enhancing its feature extraction capability and overall classification accuracy.

J. Influence of the Few Sample Condition

In this section, a limited number of instances were randomly sampled from four datasets, with each class having sampling sizes set to 5 and 10 to assess the model's robustness and generalization capability. The experimental results are presented in Table XIII, which demonstrates that various comparative algorithms experience further declines in accuracy as the sample size decreases.

Specifically, compared to CNN-based algorithms such as 2DCNN and TriCNN, which exhibit significant decreases in accuracy on the UP dataset, SAGFFNet, leveraging the foundation model, demonstrates enhanced feature extraction capability and sensitivity to small sample sizes. This sensitivity enables SAGFFNet to achieve classification accuracy that significantly outperforms other comparative algorithms under limited sample conditions.

K. Analysis of Runtime and Network Parameters

In this section, the performance parameters of each algorithm are compared, e.g., training time, testing time, and the number of network parameters (Params), as shown in Table XIV. It can be seen that the training and testing times of the algorithm are at a moderate level compared to various other methods. The network parameters of SAGFFNet are more numerous. This may be due to the combination of multilayer convolutions and the stacking of high-dimensional features.

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However, the larger number of network training parameters does not drag down the algorithm's running time; on the contrary, SAGFFNet achieves higher classification performance with comparable running time compared to the rest of the algorithms. In the future, we will attempt to explore the lightweight model, thereby further enhancing its potential for application in the field of HSI classification.

IV. Discussion

The statistical test results from comparative experiments across four datasets demonstrate that the proposed method significantly enhances the performance of HSI classification. Ablation studies were conducted to validate the efficacy of the adaptive global feature extraction module and the spectral prompt module. By leveraging a self-adaptive padding mechanism, the algorithm effectively mines spatial information between different pixels, thereby further improving the model's capability to distinguish among various categories. The spectral prompt module guides the network in classification by leveraging multiscale spectral texture information. Moreover, comparative experiments with varying sample sizes further validate the robustness of the algorithm.

Compared to existing methods, the SAGFFNet leverages the perceptual capabilities of the foundation model, achieving superior results in metrics OA, AA, and Kappa. With its precise pixel-level classification, it can be further extended to high-precision scenarios such as semantic segmentation in remote sensing tasks.

While the method delivers satisfactory classification performance, there is still room for improvement in model parameters and training time. Compared to the commonly used transformer-based GSCVIT, the SAGFFNet has longer training times and larger parameter counts. For instance, on the UP dataset, SAGFFNet's training time is approximately 99.77 s, with a trainable parameter count of around 1.74 M, whereas GSCVIT has a training time of around 47.75 s with a parameter count of around 0.09 M. Therefore, future applications could further develop toward reducing the computational complexity of the model.

V. Conclusion

A novel SAGFFNet model is presented in this article, which encompasses GFEsNet, SFPsNet, and SIDsNet. Initially, the raw hyperspectral data are dimensionally reduced to 3 using PCA, and the reduced data are segmented into multiple patches by the self-adaptive padding mechanism. While GFEsNet preprocesses these patches to extract global features, SFPsNet extracts the spectral information of the original hyperspectral data and obtains the local features of the labeled samples. Subsequently, SIDsNet utilizes a convolutional structure to relearn category semantic information from the above features.

SAGFFNet has made the first attempt to utilize the foundation model to accomplish HSIC tasks. By introducing the self-adaptive padding mechanism, it validates the effectiveness of the foundation model in extracting spectral features while also demonstrating their potential in few- or zero-shot HSIC tasks. Experimental results show that the SAGFFNet is able to obtain optimal classification performance on four HSI datasets compared with the other nine state-of-the-art algorithms. In the future, we plan to leverage the framework presented in this article for open-set classification of HSI, enhance classification performance through advanced techniques, and investigate lighter-weight networks to simplify the architecture.

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